

Math 141, S06

Quiz 8

Section:

Name: *key*

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (5 points) Find $\frac{d}{dx} [e^{x-e^{4x}}]$.

$$\begin{aligned}\text{Solution: } \frac{d}{dx} [e^{x-e^{4x}}] &= e^{x-e^{4x}} \frac{d}{dx} [x - e^{4x}] \\ &= e^{x-4x} \cdot (1 - 4e^{4x})\end{aligned}$$

Problem 2. (5 points) Find $\frac{dy}{dx}$ using logarithmic differentiation if $y = (x^2 + 3)^{\tan x}$.

$$\text{Solution: } y = (x^2 + 3)^{\tan x}$$

$$\Rightarrow \ln y = \tan x \ln(x^2 + 3)$$

$$\frac{d}{dx} [\ln y] = \frac{d}{dx} [\tan x \ln(x^2 + 3)]$$

$$\frac{1}{y} \frac{dy}{dx} = \sec^2 x \ln(x^2 + 3) + \tan x \cdot \left(\frac{2x}{x^2 + 3} \right)$$

$$\frac{dy}{dx} = (x^2 + 3)^{\tan x} \cdot \left[\sec^2 x \ln(x^2 + 3) + \frac{2x \tan x}{x^2 + 3} \right]$$